S09 local-first verification operations

Current completion and evidence boundary (2026-10-11)

Rust adoption is settled by AK; the recorded task direction also documents the local-first scope and zero-based amendment restart. Complete the scoped local/Linux-CI checks and resolve Required PR findings; S09 live measurements do not block S10/S13 implementation. Existing route goldens, correctness, placement and cutover gates remain in force. The governing issue and workflow define this scope.

The default loopback configuration covers cold, warm and simulated provider Data/NoData/denied/error at 8/16/32/64 offered clients with at most 16 owners. Overload is an observed outcome, not a reason to hide samples or raise limits. The report records foreground throughput, request percentiles/sample limitations, process CPU/RSS, fixture counts/bytes and S3-operation/catalog-version cost inputs. Prices are unset: counts/bytes are inputs for a later dated cost calculation, not invoiced cost or actual S3 physical storage. Local peers do not prove AWS latency, authentication, production durability or weather-route parity.

In the native reference, `provider_data_16` had 14 successes and two `other_error` outcomes at about one second despite 16 owner slots. Their precise cause is unclassified; neither is treated as admission overload or discarded. Both failed clients had `non_success_storage_checks: verified_empty`, while successful clients had fresh-S3-only readback checks. This is a retained local failure observation, not an all-success or latency acceptance claim. S09 releases implementation work based on checked behavior and truthful limits; actual route performance acceptance remains pre-cutover.

Linux artifact checks before EC2

The existing Server2 foundation CI installs the musl build prerequisites and builds the locked x86_64-unknown-linux-musl release target. It executes that artifact's real loopback health/startup smoke and the synthetic benchmark matrix. The regular workspace tests, GNU smoke, placement and isolated operational checks remain. Missing isolation fails closed; never run operational unittest modules directly on the host. macOS can run Rust/loopback checks but cannot claim the Linux-only isolated operational or musl execution results.

Docker preflight using the CI artifact

Run the existing Linux foundation CI first, then download the successful run's `server2-linux-musl-verified` artifact. Check out the corresponding PR source: verification scripts, fixtures and binaries must match its manifest. The native Linux checks and Docker use the same `tools/verify_artifact.py` entrypoint. No local rebuild is required when the CI bundle is available.

```
gh run download RUN_ID --repo WizardFactory/TodayWeather \
  --name server2-linux-musl-verified --dir reports/s09-ci-bundle
python3 server2/tools/verify_docker.py \
  --bundle reports/s09-ci-bundle --output reports/s09-docker-check
```

Use a fresh download/output directory for each run. Missing Docker, incompatible ELF, changed source/binary hashes or failed checks stop verification. The runner retains startup/matrix logs and a result receipt; only complete success exports a new successful manifest and both unchanged binaries. Inspect matrix outcomes: a completed verification does not mean every offered request succeeded.

On this Apple Silicon workstation, Docker29.9.0 and Colima0.10.3 are installed; the dedicated `s09-preflight` profile uses 2 CPUs/4 GiB and VZ/Rosetta, without a login service or changing the default Docker context. Its shared staging root is `/private/tmp/server2-s09-docker`. To reuse that existing profile, add `--docker-context colima-s09-preflight --staging-root /private/tmp/server2-s09-docker` to the verification command; start/stop it with `colima start s09-preflight --activate=false` and `colima stop s09-preflight`. Other hosts can use their existing Docker runtime and a Docker-visible staging root.

The launcher copies only the allowlisted verification sources and prebuilt binaries into temporary staging. The runtime container uses a digest-pinned Python image, linux/amd64, a nonroot UID, read-only sources/root filesystem, dropped capabilities, no external network and no forwarded account credentials. Only temporary/output directories are writable. An interrupted/timed-out launch attempts removal of its unique task-owned container; cleanup uncertainty cannot produce PASS.

Image download and Linux build dependency acquisition need network; the verification container itself does not. Existing operational namespace/audit tests remain native Linux CI, with no privileged-container fallback.

Docker confirms executable compatibility and bounded synthetic behavior. It does not reproduce EC2 cloud-init/systemd, SSM, IAM or real S3. Apple Silicon emulation and container limits affect timings; these observations are not EC2 performance acceptance. CI bundles expire after seven days and are disposable; maintainers retain required revision/hash/outcome summaries in the PR for durable handoff. Future EC2 smoke consumes the same verified binary hashes.

Minimal preproduction EC2 smoke (pending)

Under separate scoped resource authority, transfer the already verified musl artifact to the intended host. Verify artifact hash, runtime resources, startup, health, TLS and credentials. In a tiny isolated synthetic S3 scope, perform the required GET/HEAD/PUT/LIST operations and exact body readback. Preserve results and verify cleanup of only task-owned temporary compute/IAM resources; follow the approved S3 retention policy. No on-host build, concurrency sweep or actual provider request is part of this smoke. A failure is recorded without automatic relaunch or infrastructure expansion.

Actual implemented API latency and serving/lifecycle costs are pre-cutover checks. Raw packs remain disabled unless actual route measurements demonstrate that raw GET fan-out prevents meeting an agreed latency/request-cost target. This document grants no AWS execution, budget reset, deployment or merge.

Run an isolated release measurement

From the repository root, build the single benchmark binary. Rust workspace files and benchmark implementation stay under server2. Python 3.9 or newer is required; the actual interpreter and host are recorded in the maintained evidence manifest.

```
cargo build --locked --manifest-path server2/Cargo.toml \
  --release --bin server2-feasibility
python3 server2/tools/benchmark/run.py \
  --config server2/config/benchmarks/local.json \
  --output reports/s09-local.json
```

The 2026-10-11 native reference used macOS 26.6.2 arm64, Apple M4, 10 logical CPUs and 16 GiB physical RAM (17,179,869,184 bytes), Python 3.9.6, and rustc 1.99.0 (b940084d7, LLVM 23.1.1, aarch64-apple-darwin). CPU/RAM were observed with `sysctl -n hw.memsize machdep.cpu.brand_string`; compiler identity with `rustc --version --verbose`. The actual command in the capture used `--timeout 600`. Its 24 cases retained 720 offered-request outcomes, completed in 31.36 seconds, and reached 51,707,904 bytes peak process RSS. These single-trial observations are local reference evidence, with the actual binary/configuration/source hashes and measured summaries in the maintained evidence manifest.

The maintained runner creates a fresh literal 127.0.0.1 peer and rewrites both endpoints in an effective configuration. The template's port 1 is a placeholder; the direct CLI does not discover or start peers. `--validate-config PATH` checks the complete schema before I/O. Remote/DNS/auth/query/fragment endpoints and unknown fields fail closed. Only fixed dummy local credentials are used; profiles and environment keys are not loaded. The peer checks protocol shape, not AWS authentication.

The runner validates the requested configuration from one byte snapshot. It records that hash and the rewritten effective-config hash separately. Binary, lock, driver, model, runner and peer hashes are captured before execution and checked afterward; changed files invalidate the run. A dirty checkout revision does not assert that current source bytes belong to HEAD. The runner limits process duration to 1..600 seconds (default 300), kills only its owned process group on timeout and rejects output above 8 MiB. Keep output under ignored reports; it is an observation, not source.

Choose a valid workload

Each case specifies name, workload, mode, selection, clients, owners, trials, revisions and record_bytes. Selection is targeted/latest/full_history; offered clients are 1..64, owner admission 1..16, revisions 1..240, trials 1..100 and body size 32..65536 bytes. Optional siblings=2 and layout=orphan/partial exercise record/group recovery. Revision stress uses valid immutable singleton groups; counts 30/60/100/160/240 are configurable, not automatically claimed measured.

`history8` uses eight actual day catalogs, 24 hourly identities per day and 192 singleton groups. Existing request period limits require 16 scopes (16+8 hours per day). It is a synthetic 16-resolution batch, not one existing API. Any component failure makes the batch unsuccessful; all component outcomes remain visible. No missing legacy history is fabricated.

Aggregate limits are 32 cases, 128 trials, 4096 offered samples, 2048 groups, 32 MiB fixture raw bodies and a conservative 128 MiB catalog-version estimate. Thus history8 with 16/32/64 clients is rejected before I/O. Smaller revision workloads support higher offered client counts. Validate each proposed matrix; do not silently reduce revisions or increase production limits to obtain a result. Workload generation itself may hit existing bounds; a setup failure yields no successful report.

Read durations and failures correctly

Principal all-request p50/p95/p99 uses nearest-rank integer microseconds for every attempted success or failure. A separate success-only distribution has its own denominator. Warm-unavailable entries have null duration, never zero; both distributions flag fewer than 100 measured samples as insufficient. Each default case has 8, 16, 32 or 64 offered clients in one trial and cannot establish a statistically supported latency tail.

Local runner trials use fresh resolvers. The approved-driver warm case instead reuses one resolver across observation waves, with one complete prewarm and maintenance idle; cold trials still start with empty resolver memory. Warm prerequisites must all succeed before counters are sampled. A warm report with idle baseline must have zero wire I/O. Foreground and background maintenance counts are separate. Capacity, Timeout and other typed outcomes remain visible; they cannot become a partial success or provider miss.

Offered clients are distinct keys, not parallel raw reads. The owner deadline remains 3 s, logical cache 128 MiB and owner reservation 64 MiB. View CPU=2; catalog/raw libraries each have their own bounded I/O=16 and CPU=2 pools. These are not a process RSS limit. S06 descriptor normalization is sequential; FullHistory page-size 1 repeats normalization and can hit page/deadline bounds. No optimization or old parallel-wave latency claim is made.

Client adapters record elapsed time, received status and known request/response body sizes. Cancellation leaves received status unknown. The first 4096 completion samples per operation are retained; a truncated distribution does not prove the full tail. All-call counters remain. Peer service durations use a separate capped 20000-sample distribution, not network RTT. Counters include LIST, status/conflict outcomes, bytes, peak in-flight requests and catalog current/version body bytes. Local version accounting is not AWS billing/conformance proof.

Warmup view timings are separate from foreground parse/assembly totals. Process-specific wait4 CPU and peak RSS include startup/seeding; cold request quantiles do not. Darwin RSS is bytes; Linux KiB is converted to bytes; unknown units remain null. Parse/assembly spans are elapsed wall time inside the view pool, not per-thread CPU. Seed, foreground, maintenance and fresh durability checks are recorded separately. Costs/prices remain null until supplied with provenance.

Funded miss and durability checks

Provider modes run real local S08 reservation/executor code through a benchmark-only adapter. Data receives raw+descriptor+catalog publication before client success, followed by a fresh S3-only lookup that checks exact bytes. NoData/code 03 remains RAM-only and terminal. Funding denial and provider errors cannot archive weather raw/group/catalog objects; their zero PUT fence is validated independently of allowed budget writes. Denial makes zero provider calls. This fixture seam does not deliver a production S07/S08 provider adapter or weather semantics.

Every local report retains local_evidence_only_preproduction_checks_pending, adopted_by_AK, route_implementation_requires_live_measurements=false and ec2_smoke_status=pending_preproduction. API parity and production cutover remain false. Packs remain disabled. The local CLI has no live fallback; retained AWS tooling needs separate current authority.

Verification and references

Regression tests cover failure-inclusive quantiles, null/unavailable denominators, RSS units, target/budget bounds, cold/idle-warm HTTP, orphan siblings, actual eight-day identities, admission pressure and funded Data/NoData/denial/error fences. The distinct release runner supplies the usage capture below. Its low-sample results demonstrate tool behavior only. Existing S05-S08 tests and placement/artifact checks remain required.

See the architecture, task specification. The separately linked root execution/cleanup observation below is an actual control-plane attempt, not a successful benchmark.

Historical campaign appendix

The following revision-bound grants, commands and outcomes preserve the previous campaign. They do not define the current minimal EC2 smoke or renew execution authority. Former Rust/O-2 and live-measurement blockers are superseded above.

Historical delivery and AWS campaign records

Historical status: the maintained successor integrates offline startup/lifecycle corrections; the Q9 contract below covers selected guard retention, cleanup diagnostics and Linux verification corrections. Review and test evidence always applies to its recorded revision. This is not a new live measurement or an execution receipt. The historical attempts and reviews below apply only to their original bytes; actual intended-host measurements remain incomplete.

This tool exercises the actual Rust raw store, catalog, resolver and funded-acquisition libraries against fresh volatile HTTP peers. It measures synthetic origin work, not a deployed weather route. The local runner calls no real AWS/providers. A separately armed approved driver can use one dedicated same-region test bucket after eligible review. The approved recovery attempted host creation once but stopped on the console-output boundary before any worker allocation or measurement. Root subsequently verified owned cleanup. This proves no API parity, CloudFront latency, measurement-path IAM/TLS or intended-host musl feasibility. S09 remains open: same-region intended-host measurements and O-2/O-5 decisions still block S10/S13.

Separately authorized sixth attempt (2026-10-10; historical)

After the private-manifest ancestor repair, AK explicitly approved one additional AWS attempt, at most USD5, cumulative six attempts/USD30, with fixed expiry **1791648000000 (2026-10-10 16:00 UTC / 2026-10-11 01:00 KST)**. Publish the minimum successor, then execute it once; no automatic retry or merge. Earlier approvals and failed states remain historical, unchanged and non-refundable.

The fifth host built successfully but the worker rejected its private manifest before a valid report. Its issued worker1 allocation stays consumed in retained history; this new approval funds a distinct worker2, not a resumed/refunded worker1. Keep per-worker request/byte/storage limits, per-host7200s, benchmark window, export/cleanup tail and controller caps unchanged. The new worker allowance is separate from the predecessor's retained full reservation, not a reset of usage.

Before creating state/watchdog, require the immutable cleaned fifth cumulative state and exact bound cleanup/failure attestation, including predecessor1-4, reconciliation and fourth-cleanup pins. Debit fifth cumulative controller usage once plus its full watchdog50 and conservative external100 reservations. Reject missing/mismatched/regressed/exhausted inputs and arbitrary or uncertain worker outcomes. Preserve original state and artifacts. Fresh resource absence, final published source pins and actual CLI wire limits remain launch prerequisites. Collect bounded evidence before ordinary cleanup; preserve S3 versions. This approval is not a measurement-success claim.

Fifth dispatch authorization (2026-10-10; historical)

After the bootstrap diagnostics correction, AK authorized commit/push and one additional AWS execution. Explicit confirmation grants one additional attempt with USD5 maximum, cumulative five attempts/USD25, expiring at **2026-10-10 14:30 UTC (23:30 KST), epoch1791642600000**. No automatic retry or merge is authorized. The prior four consumed reservations remain consumed; these are ceilings, not claims about invoiced spend. This is a new authorization, not a rewrite of the 11:18 UTC fourth-attempt receipt. Per-host7200s, benchmark3600s, API/byte/storage caps and export/cleanup reserves remain unchanged.

The fifth controller requires the immutable fourth cleaned pre-worker cumulative state and a hash-bound cleanup attestation in addition to the original two states and third-state reconciliation. Carry the fourth cumulative ledger once, verify its predecessor pins and non-regressing counters, and charge its watchdog plus conservative external verification reservations. Missing, changed, worker-issued, unverified-cleanup or exhausted inputs fail before state/watchdog creation.

Build and execute the exact published PR head, not local overlays. Newly renewed Python/Rust/host/config expiry bytes and source pins require fresh verification. Final-config user-data must pass the installed CLI wire check before the single launch reservation. Preserve bounded failure evidence before ordinary cleanup; retain S3 versions. Authorization alone is not an execution or measurement result.

Bootstrap failure diagnosis and retained evidence (2026-10-10)

The fourth dispatch created its host and passed guard, SSM Online and monotonic admission, but the bootstrap SSM command returned exit 1. The retained invocation reports `PT0.118S`; retained console identifies Ubuntu 26.04.1 and contains `S09_BOOTSTRAP_FAILED`. The private bootstrap log was not exported before ordinary termination. This is not evidence of a Rust compiler failure or successful measurement. Host, root volume and owned IAM/profile/SG cleanup were verified; S3 remains retained. No further dispatch is authorized by this diagnosis.

The baseline APT rewrite rejected `http://` anywhere in `ubuntu.sources`, including documentation comments. The cloud-init 26.1 upstream template contains `# See http://help.ubuntu.com/community/UpgradeNotes`. The exact baseline fragment fails on an upstream-derived fixture; removing that comment alone makes it pass. This is a **reproduced source defect and the strongest explanation**, not proof of the lost host file or traceback. Other early file/signing/marker errors remain possible. Current template/image manifests are not the historical AMI's package attestation.

The rewrite now distinguishes active deb822 fields from comments, selects the security endpoint from the `Suites` field, requires the fixed HTTPS URI allowlist including continuation lines, and requires the Ubuntu signing key per active stanza before writing. Documentation comments are preserved; unsupported active URIs or missing signing fields fail closed without partially changing the file.

Bootstrap evidence must be returned within the original SSM command before its terminal error triggers ordinary cleanup. The command retains its original exit code and reserves diagnostic time inside, not beyond, the existing 2100-second bound. Explicit stage markers separate guards, APT source preparation/update/install, Rust installation, source verification, build and binary verification. Bounded reads of only the fixed private bootstrap/build logs produce allowlisted error signals and counts/digests of withheld content—not full raw log copies. Controller records retain command identity/status/code and the sanitized diagnostic artifact before termination. Malformed/missing output or a failed local evidence write is reported unavailable; it must not replace the original failure or prevent cleanup. Emergency expiry/quota shutdown can still preempt reporting and is never delayed.

User-data correction and separately approved fourth attempt (2026-10-10)

The third dispatch was rejected by EC2 at 09:33:07Z with `Client.InvalidParameterValue: Encoded User data is limited to 25600 bytes`. Its same-client-token CloudTrail event is `93fea912-2f69-40e5-a387-96b0b8c9a602`. No host or worker was created; exact owned temporary IAM/profile/security-group cleanup and empty instance/volume inventories were read back. The retained S3 bucket had no versions. The controller's original unresolved state is preserved; service rejection and subsequent cleanup are separate evidence, not rewritten controller success.

AK separately approved **one additional attempt, at most USD5**, for cumulative limits of **4 attempts/USD20**, with the unchanged **11:18:00Z** cutoff. This is not a refund of the third dispatch or a reset of prior request/byte reservations. No automatic retry, replacement worker, production mutation or merge is allowed.

For this fourth attempt the operator requires both original `--prior-state PATH` inputs plus `--reconciled-prior-state PATH` (the immutable third cumulative state) and `--prior-reconciliation PATH` (the state/token/config-bound service rejection and owned-cleanup attestation with evidence hashes). It checks that the third state already includes the first two, then carries that total **once**, adds its full 50-call watchdog reservation and conservatively reserves 100 external reconciliation calls, reads, writes and bounded responses. Missing/mismatched proof, claimed worker, regressed/exhausted accounting or recovery state fails before creating new state. Original UNKNOWN and all input bytes remain unchanged. The receipt does not replace fresh AWS absence checks or campaign authorization.

The gzip payload is supplied to AWS CLI as a private binary `fileb://` input, not as an already base64-encoded argument. The EC2 CLI customization performs the one required encoding. Both decoded (16,384 bytes) and encoded (25,600 bytes) limits must hold before dispatch. Installed-CLI verification uses an isolated loopback EC2 stub with no credentials or external network; one base64 decode of its captured wire value must equal the exact gzip bytes. This proves transport, not cloud-init execution, IAM readiness or completed AWS measurement.

Maintained startup successor (2026-10-10; historical three-attempt authority)

AK explicitly renewed the remaining single PR-branch attempt to fixed expiry **2026-10-10T11:18:00Z (1791631080000)**. The preceding 09:18:00Z / 1791623880000 renewal and original 04:49:59Z / 1791607799000 approval remain historical references, not additional grants. Supplied authority records two of three host attempts consumed and USD 5 remaining of USD 15 cumulative; this code update neither rechecks live inventory nor resets attempts, spent usage, S3 versions, recovery allowance or worker claims. The per-host 7200-second cap, worker windows, export reserve and all request/byte grants are unchanged. An expiry renewal alone never authorizes a new launch or a replacement worker. For the explicitly approved final attempt, supply both original cleaned pre-worker files using repeated `--prior-state PATH`. The operator pins their canonical bytes and debits their spent ledgers/control calls plus full prior watchdog reservations before dispatch. Inputs are never modified; exhausted, claimed, uncleaned, recovery or already-carried states fail closed. This input is accounting only, not live absence or campaign authority. The campaign still reserves exactly one final host and USD 5 without resetting its earlier totals.

The controller read-validates any retained bucket's fixed owner, region, versioning, public-access block, AES256 policy and exact bucket policy, then inventories all versions without refunding usage. It never rewrites settings on reuse. New instance-profile membership receives at least 60 seconds of settling/readback within a 120-second bound before the single launch attempt. Neither IAM visibility nor the following readiness probe proves credential issuance or SSM authentication.

Final-stage cloud-init keeps SSM stopped until both fixed-endpoint TLS probes and IMDSv2 iam/info Success plus the expected role listing are ready. Recovery never requests a credential body. Fresh expiry/quota guards fence restart. Within the 180-second readiness cap the host captures readiness wall/monotonic time before publication or restart. A private per-launch HMAC binds the proof to the nonce, run, source, role and deadline. **Online admission is strictly [readiness, readiness + 300 seconds)**, clipped by absolute expiry; delivery, restart, polling and handshake latency consume that same window. Missing, forged, replayed, future or out-of-order proof fails closed. A control-only SSM handshake independently checks host monotonic age and clock consistency. Only its bound host acknowledgement permits the separate build command; an existing bootstrap.sh is not admission. The supervisor is 510 seconds plus a 2-second kill grace, never an extension of absolute expiry.

Ordinary recovery/bootstrap/worker failures return bounded diagnostics or preserve results before one controller-owned final cleanup phase. Normal termination dispatch is persisted once, with bounded reconciliation polls. Independent expiry/quota/orphan-watchdog emergency protection stays armed and can preempt collection. S3 objects/versions are always retained.

The recovery script is maintained beside cloud-init, not a runtime overlay of old source bytes. Its exact bytes are included in the task source map and embedded before network admission. Python and Rust require the exact unique path set in server2/config/tasks/S09.json, rather than a historical count. Cloud-init still checks out the exact reviewed revision, checks every mapped hash and builds that head. Regenerate source/config/review pins for the final committed tree and obtain a fresh independent review. No prior receipt or scratch override manifest validates this successor; checked-in pins stay inert. Do not repin or rewrite the original measurement snapshots and receipts.

Linux Python operational tests **must** use the maintained runner below. Bubblewrap, working user namespaces and /usr/bin/python3 are prerequisites. The runner clears environment, drops capabilities, isolates PID/network/mount namespaces, hides host services/sysfs/credentials, mounts source read-only and permits only reviewed fake-aws children beneath private temporary scratch. It verifies denial of an inert command before imports. Missing isolation is a hard failure, including in CI; never run the operational tests directly or substitute a host-side fallback. All functional outcomes are fake-only.

```
python3 server2/deploy/benchmark/run_isolated.py safety
python3 server2/deploy/benchmark/run_isolated.py 'test_*.py'
python3 server2/deploy/benchmark/run_isolated.py functional
python3 server2/deploy/benchmark/run_isolated.py imds-functional
python3 server2/deploy/benchmark/run_isolated.py online-functional
```

The existing architecture diagram remains a revision-bound historical measurement/recovery overview, not evidence for the maintained successor bytes. No route, collector, response, storage or deployed topology changes in this correction. The detailed admission/cleanup contract is maintained here; no newly validated diagram, live readiness or host-musl proof is claimed.

Q9 guard evidence, cleanup failures and local verification

The current declaration contains 40 server2 paths plus 11 outside documentation paths (51 total). The source map follows that exact set, not an old 33-path count. Intent/spec/plan distinguish historical corrections from current requirements. The current Q9 plan declares its exact paths before dependent edits; no independently retained d95 pre-edit declaration was available in this worktree, so that historical evidence gap remains explicit.

After the controller validates the initial guard and opens HTTPS, it retains the proof and digest with the instance and verification time. A later signed readiness record does not need the old guard line to remain in the latest console snapshot: the controller revalidates the preserved proof and current run/nonce/source/deadline/instance/time binding. A guard still present in the latest snapshot must validate and precede readiness. Latest failure markers and malformed/conflicting guards always reject, even with a good stored proof. Missing or invalid stored evidence cannot authorize admission. Existing HMAC, readiness/Online clocks, monotonic handshake and subsequent stock-guard checks still apply; no extra dispatch or time grant is added.

If work and final cleanup both fail, the operator preserves the original work exception as the primary failure. State separately records `lifecycle_cleanup_failure` with `work_failed`, `cleanup_outcome=unverified` and `watchdog_remains_armed=true`; the chained cleanup diagnostic is sanitized. State persistence is best effort if the state write itself fails; the in-memory diagnostic and original exception remain, and no durable write is claimed. Successful work with cleanup failure still fails. Check the retained state for cleanup outcome rather than treating a worker error or termination request as verified host absence. One final cleanup phase, bounded polls, once-only ordinary termination and independent emergency protection remain unchanged.

Full `cd server2 && bash tools/ci.sh` verification is Linux-only. It requires Python3.11+ for the placement checker, /usr/bin/python3 for the sandbox, bubblewrap, working user namespaces and the declared Rust toolchain. GitHub Actions prepares bubblewrap/AppArmor and loads the pinned upstream bwrap profile when the host restricts user namespaces. On macOS, syntax/source/JSON/placement checks remain useful but do not exercise operational isolation; run the full harness in Linux CI. Never execute the operational unittest files directly, install a host fallback, or disable global namespace restrictions to obtain a pass.

The c44e9e1a baseline CI passed 124 isolated operational tests and fake-only functional checks; that count is a dated baseline, not a fixed suite size or proof of later bytes. Q9 adds regression cases for guard scroll-out, invalid cache/latest failure refusal and work/cleanup failure precedence. Its final test results must be tied to the final PR head. The existing local workload screenshot remains historical and is reused unchanged when this manual's PDF is refreshed. No new benchmark capture, AWS/SSM/provider call or live readiness is implied. R5 is outside this correction.

Approved driver and operator boundary

The original resource approval specified the historical 2026-10-10T04:49:59Z cutoff (superseded only by the fixed renewal above). That historical grant permitted one dedicated Seoul c6i.large host for at most two hours, one hour of benchmark work and an operator stop at USD 5. No production hosts, actual weather/geocoder providers, MongoDB or notification state are used. The fixed account is 141248341265; the bucket and temporary role names are fixed by the inert templates. Synthetic S3 objects and all their versions remain after cleanup. Owned host/root storage, temporary IAM and security-group cleanup never delete bucket objects.

These validation commands perform no network I/O:

```
server2/target/release/server2-feasibility \
  --validate-aws-config server2/config/benchmarks/aws.json
python3 server2/deploy/benchmark/aws_operator.py
```

The operator template is inert. Actual execution requires reviewed source/config/lock hashes, actual host-musl binary hash, an eligible native review receipt, explicit authorization and a fresh private state file for initial creation. The operator first creates a host with no egress, proves the stock timer/kernel guard, then permits outbound IPv4 HTTPS to 0.0.0.0/0:443 for the entire run. This is a port restriction, not an AWS destination allowlist; it does not authorize arbitrary workloads. It verifies an Online SSM agent with a strictly parsed AgentVersion of at least 3.3.40.0 before the first RunCommand, build hashes and the actual getconf CLK_TCK value. A private source-bound manifest grants one allocation; worker.claim is exclusive and cannot be resumed. A lost worker has cleanup authority only. Operator calls, IMDS, SSM/log export and bootstrap traffic have separate bounded grants.

The live binary entry requires --aws-config, --run-manifest and --execute-approved-run together. It refuses arbitrary endpoints and requires the fixed private manifest location, Linux x86_64 musl, source/binary/config hashes and the actual role identity.

IMDSv2 credentials stay in RAM and refresh coherently; no profile discovery, secret output or URL logging is allowed. Protocol preflight uses actual HTTPS adapters, hash/MD5/gzip checks, 404, conditional 200/412 and CAS/version checks. Signed missing-condition and forbidden-namespace probes must return 403 before any measured sweep. None of those real-host proofs is asserted by local tests.

Attempt, byte and version accounting

Worker limits are 390,000 GET/HEAD, 18,000 PUT/LIST, 26 GiB reserved download and 63 MiB stored-version charge. Controller limits are 10,000 GET/HEAD, 2,000 PUT/LIST, 3 GiB bootstrap plus 1 GiB administration and 1 MiB store. IMDS uses at most 512 MiB of controller administration, not an additional grant. Every potential PUT permanently charges body length plus 16 KiB, including failed, denied, duplicate and ambiguous attempts. This allowance is conservative bookkeeping, not exact billing. No refund or object inventory renews it.

Read phase caps are protocol 5k, cold 280k, repair 60k, warm 10k, funding 10k and failure 25k. Every delegated retry/CAS/LIST/repair/budget operation receives a ticket before dispatch. Credential waits and local validation may make admitted attempts an upper bound rather than exact received requests. Confirmed status/body sizes, reserved bytes and unknown completion are separate. Unknown S3 or metadata work fails overall compliance. The controller independently checks metadata requests at most 1,024, bytes at most 512 MiB, unknown_calls=0 and worker overall compliance.

Canonical fixture estimates include gzip, descriptor and every intermediate catalog version. Read limits use the largest generated raw/control body plus 1,024 bytes of margin, within existing runtime bounds. The estimate can refuse a case; actual tickets remain authoritative. A read one byte above its cap cannot certify accounting. Real NIC arrivals/retransmissions are not physically bounded by application counters; stock guards and operator observations provide separate backstops.

Interpret the approved-driver report

The bounded 32-case template covers cold/warm/targeted/latest/full-history, a synthetic eight-day batch, one first-flight orphan repair and local funded Data/NoData/denial/error behavior at selected offered levels. It is not a full Cartesian matrix. Reuse immutable physical fixtures across fresh cold trials; report groups, revisions, day catalogs, resolution count, response keys and offered/admitted clients separately. Repair samples describe one initial orphan wave, not later healthy-catalog requests as repair p95.

The target_samples field counts offered client observations. independent_RAM_empty_resolver_trials counts actual fresh resolver trials and is zero for warm cases. warm_observation_waves counts repeated waves through the one retained warm resolver. independent_trial_tail_insufficient uses fresh resolver trials, so it remains true for warm cases; warm_wave_tail_insufficient separately describes the warm-wave population (null for other modes). Shared waiters are not independent backend samples. The current template does not prove the original 300 independent cold trials per level; omitted/refused or underpowered coverage is INSUFFICIENT. Warm resolver metrics are case-cumulative; ledger before/foreground/owned-work deltas isolate each wave. A verified second warm wave must add no S3/provider I/O.

Network clients are shared after protocol and seeding. RAM-empty measurements therefore include warmed TCP/TLS/DNS pools and do not model a fully fresh Spot process. Phase-tagged first protocol samples are separate. Process RSS/CPU includes protocol and seeding; Linux tick conversion needs the actual host getconf receipt. The local history8 batch took 14.054289 s in a historical fixture: this is a feasibility risk, not an existing weather API or CloudFront latency claim.

The maintained operator test can drive the compiled worker through a fake CLI/SSM controller and real loopback S3/provider peers. This functional proof uses the same source-bound manifest, claim, ledger and workload code; it asserts real AWS=0, metadata=0 and host gate=false. It cannot substitute for the approved intended-host measurement or AK O-2/O-5 decisions. S09 stays open and S10/S13 stay blocked until those original gates hold.

Fixed host reads and placement guard

The historical 33-path amendment permits four source/operation/path roles only: Path::new on /opt/server2-s09 and /opt/server2-s09/run for private ownership checks, and read_to_string on /proc/self/status and /proc/self/stat for this process's diagnostics. The gate validates the canonical S09 declaration on later whole-tree audits too. Declarations cannot add source files, other operations, prefixes/globs or compiler/encoded/legacy exceptions. Static checks remain an aid to human review, not a filesystem sandbox.

Failure reports and uncertain work

After a valid one-use worker claim, an aborted run emits a bounded JSON report on stdout and exits 2. run_status is ABORTED, abort_reason is a sanitized stable class, requested_case_coverage_complete is false and overall_accounting_compliance_pass is false. Completed cases and the current partial case retain summary denominators, typed outcomes, received-status counters, admitted attempts, bytes and permanent version charges. No later case is dispatched. Details that exceed the 2 MiB export limit are omitted explicitly while summaries remain; this omission also prevents a compliant result. No credentials, signed URLs or arbitrary provider errors are copied into abort_reason.

A configuration, source, manifest or exclusive-claim rejection before execution may exit 1 without JSON. The controller preserves bounded stderr and a missing-report diagnostic with false compliance. For a claimed run it exports at most 2 MiB of result and 64 KiB of sanitized worker errors, exclusively creates private local evidence and persists its digests and false compliance before validating worker status, identity or accounting. Rejected, failed or timed-out work never becomes a successful measurement. Cleanup runs after evidence preservation; export failures are themselves recorded, not treated as an absent failure.

IMDS accounting distinguishes confirmed non-200 status and response-header/length/stream-cap rejection (failed_known) from incomplete transport, cancellation or unknown completion (unknown_calls). Confirmed failures retain conservative reservation and status information; they never permit metadata preflight to continue. Truly unknown work remains charged and makes overall compliance false. Local controller-worker proof uses no actual IMDS, IAM, SSM service or AWS endpoints.

Host I/O inventory and guard limitations

The nonliteral inventory clarification is recorded in S09.json for manual source review. It is distinct from the unchanged four literal machine waivers. Review covers the bounded private manifest read and validated ancestry; adjacent exclusive worker.claim create/write/fsync; hashes of the exact task-declared source paths and Cargo.lock under the build checkout; bounded current_exe hash; and caller-owned configuration plus bounded private result/error/state files. The local seam uses its owned temporary inputs and ignored reports. Computed paths, concatenations and aliases are not a complete static security boundary; every nonliteral access requires manual ownership and source review. This inventory grants no arbitrary create/open/include exemption.

The stock kernel thresholds are intentionally stricter than the accounting grants: 2 GiB bootstrap, 24 GiB benchmark and 28 GiB global, compared with 3/26/30 GiB application grants. They reserve headroom for uncontrolled traffic and may drop traffic and abort a run earlier. A grant is a ceiling, not an entitlement to consume the full amount. Independent host/guard/version proof remains unexecuted until the approved live preflight succeeds.

Cost assessment and operator stop

The USD 5 operator stop is an operational authorization limit, not a dynamic billing monitor or hard invoice cap. A fresh regional price assessment precedes arming. The 2026-10-08 official rates imply known maximum components of approximately USD 0.43610055: c6i.large two hours (0.192), public IPv4 two hours (0.010), 16 GiB gp3 for two hours using a 730-hour month (0.00399781), 400,000 GET/HEAD (0.140), 20,000 PUT/LIST (0.090), and 64 MiB S3 retention for 48 hours (0.00010274). These are conservative authorized quantities, not actual usage or an invoice.

The estimate excludes taxes, unknown bootstrap/log transfers, billing granularity, delayed termination and permanent retention. Retained 64 MiB alone is approximately USD 0.0015625 per month at USD 0.025/GiB-month. Objects are not deleted to control cost. The operator must assess these gaps and stop within authority; code cannot guarantee the total bill. Rates are revision-bound observations from the official Seoul S3 regional offer, EC2 regional offer and VPC public IPv4 pricing. Actual host cost and feasibility remain pending.

Case admission and export time reserve

Before the next case's estimates, seed publication or resolver construction, prior unresolved S3 uncertainty closes admission. Completed and partial earlier outcomes and permanent charges remain in the bounded non-compliant report. Bounded reconciliation already owned by the current operation remains allowed. The two-cold-profile final-trial response-loss fixture checks that no next-case request, PUT body or stored-version charge is added.

The controller reserves 15 minutes before the earlier host or approval deadline. The effective benchmark window is at most 59 minutes 30 seconds, within the approved 60-minute ceiling, with up to 30 seconds of completion grace inside the reserve. At

least 14 minutes 30 seconds then remain for bounded best-effort export and owned cleanup. A worker window below 5 minutes is refused before arming and is checked again after transfers before dispatch. The SSM command cap stays 3,600 seconds. This reserve does not guarantee SSM availability or completed export and does not extend host life or grants.

A non-compliant abort still goes directly to conservative cleanup after preserving available failure output. Post-abort guard readback and observed version inventory may be unavailable; this is an accepted safe evidence limitation, not a compliant result or a reason to renew the allocation.

Local CLI rejection and watchdog diagnostics

A newly observed AWS CLI exit 252 with its exact Unknown options marker is classified as a sanitized local parsing rejection. Raw stderr and argument values are not published; digits in a rejected option cannot become an AWS HTTP status. Other unknown subprocess outcomes remain conservative. See the official return-code contract. This classification applies to the new observation only. An offline reproduction of an old invalid command does not recover its discarded stderr or establish the original launch's absence.

Watchdog writes owned_termination_requested only after verifying an owned host and successfully requesting termination. An attempted launch with no resolved unique host is original_launch_unresolved; no recorded launch is no_launch_recorded. Every dispatched call, including an error or lost acknowledgment, is charged before dispatch against the unchanged 50-call reservation. Call failures retry with one-second backoff; ownership is re-read before each termination attempt. Empty/multiple/foreign discovery stops without termination. Exhaustion or invalid state/response leaves a sanitized diagnostic and actual/failed call counts in the exclusive finish receipt. No exception text is copied. All three diagnostics set host_absence_verified=false. A requested termination is not root-volume disappearance or completed cleanup. Outside the newly authorized recovery path below, UNKNOWN zero/multiple discovery still blocks destructive cleanup; an empty list alone is not cleanup authority.

The historical local check was `python3 server2/deploy/benchmark/test_aws_operator.py --cleanup-functional-smoke`. That direct command is retained as a record only; use the current Linux namespace/audit runner above for operational verification. It exercises actual isolated fake CLI subprocesses, truthful watchdog states and the unchanged UNKNOWN cleanup fence. It calls no AWS/IMDS and reads no configured account profile or original private run state. Existing synthetic release workload capture remains a separate local benchmark demonstration.

The cleanup-only total 15 permission, later ordinal 11 stop and ordinal 12 PR-only resumption are historical instructions and remain recorded. AK subsequently authorized correction 13 to fix source, reconcile the original execution, recover once if necessary and clean up owned resources. The installed ten-attempt helper and original counters remain untouched; the separate same-task journal documents the explicit exception. No merge or budget reset is authorized.

Recover the original failed creation once (historical procedure)

The direct test-module and installed-CLI command lines in this historical recovery procedure and the following console section record earlier verification. **Do not rerun those commands from the current checkout.** Current operational verification uses the maintained Linux namespace/audit runner above. Actual installed-CLI or compiled-worker observation helpers require a separately reviewed isolation setup; they are outside ordinary CI and this correction.

Use --resume-failed-creation with --config, --original-config, --state, --authorization and --recovery-authority. Root prepares private inputs after exact-head CI and eligible review. Do not use --execute-reviewed-run or a fresh state to retry: they would create a new run. The original run/token/nonce and acknowledged bucket/role/profile/SG are retained. The exact recovery authority binds canonical original state/config hashes, reviewed new config/source/review, user approval receipt and root-attested quiescence of the original operator/watchdog. Wrong pins, changed resources, an issued worker allocation, exhausted usage or replay are refused.

Exclusive predecessor files with mode 0600 and recovery claim files preserve original deadline/config and UNKNOWN. The new deadline is at most two hours from retry start and the unchanged approval expiry. One additional 50-call watchdog reservation comes from the same global controller caps; original calls/download/version charges and the3 GiB bootstrap reservation are not reset. Recovery watchdog files use distinct attempt 1 names.

Current token/RunId host inventories must agree. One owned host receives cleanup only and emits an explicit no-measurement result; it never adopts an old/expired guard for new measurement. Zero hosts plus zero tagged volumes and verified original closed SG/IAM/profile/private-versioned-encrypted bucket permit exactly one --count 1 launch with the original

token. Multiple, foreign, mismatched or unknown inventories stop. IdempotentParameterMismatch never permits a replacement token or another launch. A lost response is reconciled only against the original token.

New local exit 252/mismatch or prelaunch failure does not erase historical UNKNOWN. Explicit current-absence cleanup authority permits deletion of only acknowledged IAM/profile/SG after fresh zero token/tag/volume inventories and exact ownership/dependency checks. A newly uncertain sent retry cannot use that cleanup path. S3 and every version remain retained; current cleanup is not retrospective proof that the discarded original CLI result was unsent. Root reconciled the recovery13 cleanup as described below; that timestamped observation does not establish feasibility or authorize another launch.

Historical verification commands (record only; do not rerun directly):

```
python3 server2/deploy/benchmark/test_aws_operator.py
python3 server2/deploy/benchmark/test_aws_operator.py --offline-launch-oracle /opt/homebrew/bin/aws
python3 server2/deploy/benchmark/test_aws_operator.py --recovery-functional-smoke
python3 server2/deploy/benchmark/test_aws_operator.py --recovery-failure-functional-smoke local
python3 server2/deploy/benchmark/test_aws_operator.py --recovery-failure-functional-smoke mismatch
```

The offline oracle uses temporary dummy credentials and disabled IMDS to validate complete source-generated args and bounded gzip through the installed CLI's skeleton. It proves parsing/SDK shape only, not EC2/idempotency/IAM acceptance. Functional checks use actual fake CLI subprocesses and isolated loopback peers. They verify retained usage, one launch, replay rejection and current owned cleanup while preserving original UNKNOWN, with actual AWS/IMDS0. The existing release workload image below remains a separate local workload demonstration; it is not a screenshot of real AWS recovery. S09 and intended-host/O2/O5 gates remain open.

Console-output response contract and recovery13 outcome (historical)

AWS CLI get-console-output decodes the API's base64 Output before returning its JSON string. The operator accepts that string directly, never decodes it again and never guesses an encoding. A response must be an object; absent Output means an empty console still awaiting proof, while non-string Output, lone surrogates, more than 65,536 characters or more than 131,072 UTF-8 bytes fails closed. The installed CLI can already have replaced invalid input bytes with U+FFFD; the operator cannot recover that lost distinction. These body limits are additional ceilings, not a guarantee that the CLI envelope will fit. Cli.call also caps complete serialized JSON plus stderr at 131,072 bytes. JSON escaping and metadata count toward that stricter envelope cap: even 65,536 quote characters can exceed it and be rejected before body parsing. No truncation, reservation increase or proof bypass is allowed. The existing stock guard identity, unique proof, status, deadline and readiness checks still precede egress and SSM. Decoded ordinary boot text alone is not guard proof.

Root's actual recovery13 record: one original-state launch began at 2026-10-08T13:48:27Z. The first nonempty CLI output contained ordinary Linux boot text, 65,535 UTF-8 bytes, with neither guard-success nor guard-failure marker. The obsolete second decode raised console encoding before any worker grant; this was not a verified stock-guard failure. Root completed cleanup at 13:54:50Z and the 13:56:40Z readback found the instance terminated, no tagged volumes, and the owned SG/role/profile absent. The retained bucket had zero versions. Measurements and O-2/O-5 remain outstanding. This code-only correction14 authorizes no further AWS dispatch.

The earlier 42-command offline skeleton check validated input options and SDK shapes but did not exercise response customizations. The additional maintained check uses the installed CLI against a literal loopback EC2 XML peer, temporary dummy credentials and disabled IMDS:

```
python3 server2/deploy/benchmark/test_aws_operator.py --cli-console-functional-smoke /opt/homebrew/bin/aws
```

It verifies one API-base64 response becomes plain CLI text, then checks a valid guard proof and rejects duplicate, wrong-status and wrong-identity proofs. Unit regressions include exact text/UTF-8 limits and a run_host check proving decoded boot output reaches the existing SSM version refusal. These local checks do not verify a live guard, host-musl build or S3 measurement. The workload capture below is retained historical local release evidence, not a screenshot of recovery13.

2026-10-09 actual execution and focused correction

The subsequent separately authorized sequential attempt created one dedicated Seoul c6i.large with an encrypted 16 GiB gp3 root. It stopped before guard readiness: HTTPS egress stayed closed, no SSM/bootstrap build or worker was dispatched, and no S3 objects were written. At 2026-10-09T13:06:08Z, AWS readback confirmed termination, no owned root volume/security group/IAM role/instance profile, and an empty retained versioned bucket. Normal termination stalled; the operator used the installed CLI's --skip-os-shutdown on that same owned host after retaining diagnostics. Original state remained byte-identical. This is a failed pre-measurement attempt, not a benchmark result.

The CLI returned 65,535 characters occupying 65,681 UTF-8 bytes, without replacement characters. The old 65,536-byte parser rejected this legitimate response. The parser now accepts the entire response within independent 65,536-character and 131,072-byte bounds. The CLI subprocess still caps its complete output at 131,072 bytes. Nothing is sliced or decoded again; failure markers anywhere in the accepted text precede proof parsing. Missing/conflicting proofs, identity mismatch, expiry and readiness failures still block egress.

The live log stopped after enabling the expiry timer and repeatedly showed cloud-init-network.service starting. Ubuntu cloud-init 26.1 network stage precedes sysinit.target, while systemd timers normally wait for it. The synchronous timer start inside bootcmd therefore forms a circular wait. This diagnosis combines live logs and source ordering; no live systemd job graph was obtained.

All stock guard setup, including timer, SSM and meter service starts, now runs through runcmd in the cloud-init final stage, after multi-user.target. The SG remains closed until the controller reads and validates the guard. The independent controller watchdog, absolute expiry and 110-minute uptime limit remain. A host that never reaches final stage still relies on that external watchdog and owned cleanup. Bootstrap builds the measurement binary but no longer invokes the workspace test suite, as requested by AK. Existing stock kernel guard capability checks remain part of admission.

At that historical correction, based on reviewed commit 7ec054b9, eight incomplete preparation15 files were preserved in a separate worktree and excluded. Retained-bucket successor creation, periodic proof re-emission and deadline refinements were not shipped in that correction. The later maintained successor above implements its own retained-bucket validation and readiness changes; this historical exclusion does not describe the current feature set. The actual attempt used those manually inspected controller helpers separately from its pinned measurement source; QA6 did not review them. The one-use recovery allowance remains consumed. No new run, worker allocation, budget refund or additional host is authorized by this PR update.

Verification for this correction uses the installed AWS CLI help, the retained real console response, YAML/Python/shell syntax and command-order inspection, plus document/placement/artifact checks. No test suite is added or run for this correction. At correction16 the old console byte-bound assertions still described the superseded contract; the subsequent external-review correction updates them as recorded below. Historical QA6 remains a tool/configuration review of 7ec054b9 only. The corrected bootstrap has not run on Linux or AWS; current review and live-readiness gates remain incomplete. S09, O-2/O-5 and S10/S13 remain blocked on actual measurement and decisions.

External-review correction: watchdog and overload reporting

The external review identified three required corrections. The watchdog now uses its reserved calls for bounded retries and always attempts to leave a finish diagnostic after an armed failure; local filesystem failure can still prevent that receipt. This is best-effort cleanup, not a termination guarantee or renewed launch authority.

For local provider_data, exact S3-only readback is required only for successful clients. Every non-success retains its original latency/outcome and gets a separate finite-scope storage observation: verified_empty, verified_present, indexed_not_folded, incomplete or verification_failed. Capacity alone does not prove pre-admission rejection; a later failure may leave durable data. Such data is reported, never relabeled as client success. For nonempty scopes, this observation can publish missing catalogs through bounded repair; verified_present and indexed_not_folded describe the post-repair state. These verification writes are counted after the measured interval. An admitted success with invalid exact bytes still fails correctness.

Existing operator assertions now match final-stage runcmd, one bounded cargo build with no cargo test, and the separate 65,536-character/131,072-byte body limits. No new test was added and no test suite was run for this correction. Warm waves are no longer labeled independent empty-memory trials. The direct release runner and its capture below are local protocol evidence only; watchdog retries and the live warm report change are source-reviewed, not AWS-executed.

Eight local cases completed. At clients=8/owners=1, provider_data retained success=1/capacity=7, with one exact stored body and seven verified-empty scopes. Cold also retained 1/7; baseline scenarios preserved terminal archive fences. These low-sample local observations cannot establish latency acceptance.

Actual local CLI usage

```
S09 — actual release HTTP smoke | loopback only

$ python3 server2/tools/verify_docker.py --bundle reports/sdlc/issue-2694-s09/docker/local-linux-bundle-
final --output reports/sdlc/issue-2694-s09/docker/local-container-final --docker-context
colima-s09-preflight --staging-root /private/tmp/server2-s09-docker

linux
Offline Docker preflight; fixed linux/amd64 image; no credentials forwarded
+ /usr/local/bin/python3 /work/server2/tools/smoke.py --binary /work/server2/target/x86_64-unknown-linux-
musl/release/server2
PASS invalid nonloopback metrics startup rejected
server2 public=127.0.0.1:38255 metrics=127.0.0.1:42587
GET public /health -> 200 OK; text/html; CORS *
HEAD public /health -> 200; empty body; Content-Length 2
GET public /internal/metrics -> 404
GET loopback metrics -> 200 no-store; counters: {"server2_cache_bytes": "0", "server2_cache_limit_bytes":
"16777216", "server2_cpu_inflight": "0", "server2_cpu_limit": "2", "server2_health_requests_total": "2",
"server2_inflight": "0", "server2_inflight_limit": "40", "server2_rejected_total": "0"}
PASS public socket cap 2 rejects excess and recovers after release
PASS unfinished HTTP headers closed within 1.5..3.5s at 2-second lifetime
PASS SIGTERM drained both listeners; no provider calls
PASS SIGTERM with partial headers: logged clean forced stop within 4.5..7s
+ /usr/local/bin/python3 /work/server2/tools/benchmark/run.py --binary
/work/server2/target/x86_64-unknown-linux-musl/release/server2-feasibility --output /results/run/local-
matrix.json
LOCAL ONLY; Rust adopted; preproduction host smoke and API cost/performance pending
cold_8: outcomes={'success': 8} all-p95=34984us success-p95=34984us
warm_8: outcomes={'success': 8} all-p95=519us success-p95=519us
provider_data_8: outcomes={'other_error': 2, 'success': 6} all-p95=1015654us success-p95=73315us
provider_nodata_8: outcomes={'acquisition_denied': 7, 'other_error': 1} all-p95=1007519us
success-p95=Noneus
provider_denied_8: outcomes={'acquisition_denied': 7, 'other_error': 1} all-p95=1007497us
success-p95=Noneus
provider_error_8: outcomes={'acquisition_denied': 7, 'other_error': 1} all-p95=1005941us
success-p95=Noneus
cold_16: outcomes={'other_error': 9, 'success': 7} all-p95=1014582us success-p95=14197us
warm_16: outcomes={'success': 16} all-p95=10us success-p95=10us
provider_data_16: outcomes={'other_error': 9, 'success': 7} all-p95=1005207us success-p95=78330us
provider_nodata_16: outcomes={'acquisition_denied': 7, 'other_error': 9} all-p95=1005043us
success-p95=Noneus
provider_denied_16: outcomes={'acquisition_denied': 7, 'other_error': 9} all-p95=1007419us
success-p95=Noneus
provider_error_16: outcomes={'acquisition_denied': 7, 'other_error': 9} all-p95=1012115us
success-p95=Noneus
cold_32: outcomes={'capacity': 16, 'other_error': 9, 'success': 7} all-p95=1013377us success-p95=14363us
warm_32: outcomes={'success': 32} all-p95=56us success-p95=56us
provider_data_32: outcomes={'capacity': 16, 'other_error': 10, 'success': 6} all-p95=1008395us
success-p95=68102us
provider_nodata_32: outcomes={'acquisition_denied': 7, 'capacity': 16, 'other_error': 9}
all-p95=1011259us success-p95=Noneus
provider_denied_32: outcomes={'acquisition_denied': 7, 'capacity': 16, 'other_error': 9}
all-p95=1006995us success-p95=Noneus
provider_error_32: outcomes={'acquisition_denied': 7, 'capacity': 16, 'other_error': 9} all-p95=1003560us
success-p95=Noneus
cold_64: outcomes={'capacity': 48, 'other_error': 10, 'success': 6} all-p95=1007740us success-p95=15410us
warm_64: outcomes={'success': 64} all-p95=12us success-p95=12us
provider_data_64: outcomes={'capacity': 48, 'other_error': 10, 'success': 6} all-p95=1009635us
success-p95=75232us
```

Actual local CLI usage (continued)

```
provider_nodata_64: outcomes={'acquisition_denied': 7, 'capacity': 48, 'other_error': 9}
all-p95=1009865us success-p95=Noneus
provider_denied_64: outcomes={'acquisition_denied': 7, 'capacity': 48, 'other_error': 9}
all-p95=1003485us success-p95=Noneus
provider_error_64: outcomes={'acquisition_denied': 12, 'capacity': 48, 'other_error': 4}
all-p95=1062643us success-p95=Noneus
Process peak RSS: 42176512 bytes; CPU includes startup/seed
PASS: bounded report, all outcome denominators and terminal archive fences
PASS static Linux artifacts; hashes unchanged; AWS/EC2 remains unverified
PASS Docker artifact preflight; EC2/AWS checks remain pending
```